

# Weekly Putnam

5 October 2026

## PROBLEM

Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a continuous function that is not constant, and suppose that  $\int_0^1 f(x)^k dx$  is an integer for each  $k = 1, 2, \dots, 2026$ . Prove that

$$\max_{0 \leq x \leq 1} f(x) - \min_{0 \leq x \leq 1} f(x) > 3.99.$$

**Answer:** a proof; the argument gives the stronger bound  $4^{1013/1014} = 3.9945\dots$  · **Level:** A5 · **Topic:** Real analysis

## Hints

1. The hypothesis says more than it seems to:  $\int_0^1 P(f(x)) dx$  is an integer for every polynomial  $P$  with integer coefficients and degree at most 2026. Which integrals of this kind are obviously nonnegative?
2. Arrange the numbers  $\int_0^1 f(x)^{i+j} dx$ ,  $0 \leq i, j \leq n$ , in a square matrix. It is a Gram matrix, and its entries are integers. What does that say about its determinant?
3. A Gram determinant is a product of squared distances. Bound the  $k$ th distance from above by trying one well chosen monic polynomial of degree  $k$  on the interval  $[\min f, \max f]$ : a rescaled Chebyshev polynomial.

## The key idea

The matrix of moments  $(\int_0^1 f^{i+j})_{0 \leq i, j \leq n}$  is positive definite and has integer entries, so its determinant is at least 1. The same determinant is the product of the squared distances from  $f^k$  to the span of the lower powers of  $f$ , and each distance is at most the size of the monic Chebyshev polynomial on the range of  $f$ , which is  $2(L/4)^k$  with  $L = \max f - \min f$ . If  $L$  is noticeably less than 4, this product drops below 1 once  $n$  is large. Integrality and smallness cannot both hold.

## Solution

Write  $m = \min f$ ,  $M = \max f$  and  $L = M - m$ . Since  $f$  is continuous and not constant,  $L > 0$  and  $f([0, 1]) = [m, M]$ . For continuous functions  $g, h$  on  $[0, 1]$  put

$$\langle g, h \rangle = \int_0^1 g(x)h(x) dx, \quad \|g\| = \langle g, g \rangle^{1/2}.$$

Let  $s_k = \int_0^1 f(x)^k dx$ , so that  $s_0 = 1$  and  $s_0, s_1, \dots, s_{2026}$  are integers. Fix  $n = 1013$  and consider the  $(n+1) \times (n+1)$  matrix

$$H = (s_{i+j})_{0 \leq i, j \leq n} = (\langle f^i, f^j \rangle)_{0 \leq i, j \leq n}.$$

**Step 1:**  $\det H \geq 1$ . For a real vector  $v = (v_0, \dots, v_n)$  let  $P(t) = \sum_i v_i t^i$ . Then

$$v^T H v = \left\| \sum_{i=0}^n v_i f^i \right\|^2 = \int_0^1 P(f(x))^2 dx \geq 0.$$

If this is 0, the continuous function  $P(f(x))$  vanishes identically, so  $P$  vanishes on the whole interval  $[m, M]$ . A nonzero polynomial has finitely many roots, so  $P = 0$  and  $v = 0$ . Thus  $H$  is positive definite and  $\det H > 0$ . Every entry of  $H$  is some  $s_{i+j}$  with  $i + j \leq 2n = 2026$ , hence an integer. So  $\det H$  is a positive integer, and

$$\det H \geq 1.$$

**Step 2: the determinant as a product of distances.** For  $1 \leq k \leq n$  let

$$d_k = \min\{\|P(f)\| : P \text{ a monic real polynomial of degree } k\},$$

the distance from  $f^k$  to the span of  $1, f, \dots, f^{k-1}$ .

**Lemma.**  $\det H = d_1^2 d_2^2 \cdots d_n^2$ .

*Proof.* Apply the Gram-Schmidt process to  $1, f, \dots, f^n$ , which are linearly independent by Step 1. Put  $q_0 = 1$  and let  $q_k$  be  $f^k$  minus its orthogonal projection onto  $\text{span}(1, f, \dots, f^{k-1})$ . Then  $q_k = Q_k(f)$  for a monic polynomial  $Q_k$  of degree  $k$ , the  $q_k$  are pairwise orthogonal, and  $\|q_k\| = d_k$  because the functions  $P(f)$  with  $P$  monic of degree  $k$  are exactly the functions  $f^k - g$  with  $g$  in that span. The vector  $(q_0, \dots, q_n)$  is obtained from  $(1, f, \dots, f^n)$  by a lower triangular matrix  $U$  with ones on the diagonal, so the Gram matrix of the  $q_k$  equals  $UHU^T$  and has determinant  $\det H$ . That Gram matrix is diagonal with entries  $\|q_0\|^2 = 1$  and  $\|q_k\|^2 = d_k^2$ .  $\square$

**Step 3: a good monic polynomial on  $[m, M]$ .**

**Lemma.** For every  $k \geq 1$  there is a monic polynomial  $C_k$  of degree  $k$  with  $|C_k(t)| \leq 2(L/4)^k$  for all  $t \in [m, M]$ .

*Proof.* Let  $T_k$  be the Chebyshev polynomial, defined by  $T_0(y) = 1$ ,  $T_1(y) = y$  and  $T_{k+1}(y) = 2yT_k(y) - T_{k-1}(y)$ . By induction  $T_k(\cos \theta) = \cos k\theta$ , so  $|T_k(y)| \leq 1$  for  $-1 \leq y \leq 1$ , and the leading coefficient of  $T_k$  is  $2^{k-1}$ . Put

$$C_k(t) = \frac{L^k}{2^{2k-1}} T_k\left(\frac{2t - m - M}{L}\right).$$

The leading coefficient of  $C_k$  is  $\frac{L^k}{2^{2k-1}} \cdot 2^{k-1} \cdot \frac{2^k}{L^k} = 1$ . When  $t$  runs over  $[m, M]$  the argument of  $T_k$  runs over  $[-1, 1]$ , so  $|C_k(t)| \leq L^k / 2^{2k-1} = 2(L/4)^k$ .  $\square$

Since  $f(x) \in [m, M]$  for every  $x$ , the lemma gives

$$d_k \leq \|C_k(f)\| = \left(\int_0^1 C_k(f(x))^2 dx\right)^{1/2} \leq 2\left(\frac{L}{4}\right)^k \quad (1 \leq k \leq n).$$

**Step 4: conclusion.** Combining the three steps,

$$1 \leq \det H = \prod_{k=1}^n d_k^2 \leq \prod_{k=1}^n 4\left(\frac{L}{4}\right)^{2k} = 4^n \left(\frac{L}{4}\right)^{n(n+1)}.$$

Taking  $n$ th roots,  $(L/4)^{n+1} \geq \frac{1}{4}$ , that is,

$$L \geq 4 \cdot 4^{-1/(n+1)} = 4^{n/(n+1)} = 4^{1013/1014}.$$

Finally,  $\ln 4 < 2$  and  $e^{-u} \geq 1 - u$  give

$$4^{-1/1014} = e^{-(\ln 4)/1014} \geq 1 - \frac{\ln 4}{1014} > 1 - \frac{2}{1014} > 0.998,$$

so  $L > 4 \times 0.998 = 3.992 > 3.99$ .  $\blacksquare$

**Remarks.** The same proof shows that if  $\int_0^1 f^k$  is an integer for  $1 \leq k \leq 2n$ , then  $\max f - \min f \geq 4^{n/(n+1)}$ . Already  $n = 553$ , that is, integrality for  $k \leq 1106$ , pushes the bound past 3.99. For  $n = 1$  the statement reads: if  $\int f$  and  $\int f^2$  are integers then the variance of  $f$  is at least 1, which forces  $\max f - \min f \geq 2$ .

## Why it is hard, and why it is original

**Why it is hard.** The natural first move is to look for one polynomial  $P$  with integer coefficients that is smaller than 1 in absolute value on  $[m, M]$ : then  $\int_0^1 P(f)^2$  would be an integer in  $[0, 1]$ , hence 0, and  $f$  would be constant. This works when the interval sits in a lucky place (for instance  $t(t-1)(t-2)(t-3)$  handles  $[0, 3]$ ), but for an interval of length 3.99 in an arbitrary position there is no visible way to write such a polynomial down, let alone one of degree at most 1013. The determinant avoids the search altogether. It uses integrality once, for a single number built from all the moments together, and it does not care where the interval lies. Two separate ideas are needed in sequence: an integer Gram determinant is at least 1, and the distances inside it can be bounded by testing Chebyshev polynomials.

**How sharp it is.** The barrier is 4, and it is real. For  $f(x) = 2 \cos \pi x$ ,

$$\int_0^1 (2 \cos \pi x)^k dx = \frac{1}{2\pi} \int_0^{2\pi} (e^{i\theta} + e^{-i\theta})^k d\theta = \begin{cases} \binom{k}{k/2}, & k \text{ even,} \\ 0, & k \text{ odd,} \end{cases}$$

so every power integral is an integer while  $\max f - \min f = 4$ . Hence 3.99 cannot be replaced by any number above 4, and letting  $n \rightarrow \infty$  in the Remarks shows that 4 is exactly the least possible range when all power integrals are integers. Continuity matters: the step function equal to  $-1$  on  $[0, \frac{1}{2})$  and  $1$  on  $[\frac{1}{2}, 1]$  has integer power integrals and range 2. In the proof, continuity is what makes the range an interval and the matrix  $H$  positive definite.

**Relatives.** The problem is a finite, quantitative cousin of a classical circle of results in which the length 4 plays the same role: the theorems of Hilbert and Fekete on polynomials with integer coefficients that are small on an interval (an interval shorter than 4 carries nonzero integer polynomials of arbitrarily small size), and Kronecker's theorem on algebraic integers whose conjugates all lie in  $[-2, 2]$ . Those are existence statements with no control of the degree, so they do not yield the stated bound from finitely many integrals; here the explicit constant comes out of the determinant computation. Hankel determinants of moments are a standard tool in the theory of orthogonal polynomials. I have not found the statement itself, with a continuous function, finitely many integer power integrals and an explicit bound on the range, in the competition or journal problem literature.

## Verification

- The power integrals of  $2 \cos \pi x$  were computed numerically for  $k = 1, \dots, 60$  (midpoint rule with 4096 nodes, which is exact for trigonometric polynomials of this degree) and compared with  $\binom{k}{k/2}$  for even  $k$  and 0 for odd  $k$ : all agree, with error below  $10^{-15}$  relative to  $\binom{k}{\lfloor k/2 \rfloor}$ .
- For that moment sequence the determinants  $\det(s_{i+j})_{0 \leq i, j \leq n}$  were computed in exact integer arithmetic for  $n = 0, \dots, 12$ . They equal  $2^n$ , which lies between 1 and the bound  $4^n$  given by Step 4 with  $L = 4$ .
- The inequality  $\det H \leq 4^n (L/4)^{n(n+1)}$  was tested in exact rational arithmetic for 60 random polynomials  $f$  of degree at most 4 with rational coefficients and  $n = 1, \dots, 5$  (300 determinants), and in floating point for six non-polynomial functions and  $n = 1, \dots, 6$  (36 determinants). There were no violations; the largest ratio of the two sides was 0.51.
- Arithmetic:  $4^{1013/1014} = 3.994535\dots$ , and the smallest  $n$  with  $4^n (3.99/4)^{n(n+1)} < 1$  is  $n = 553$ .
- Sharpness for  $n = 1$ : for  $f(x) = a \tanh(s(2x - 1)) / \tanh s$  with  $a$  chosen so that  $\int f^2 = 1$  (and  $\int f = 0$ ), the range  $2a$  is 2.236, 2.020 and 2.0006 for  $s = 5, 50, 500$ , approaching the bound  $4^{1/2} = 2$ .